

BrightLayer Stores - Basic Cybersecurity Risk Assessment

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FIVE IMPORTANT ASSETS

1. **Employee information**: This includes the employee email address, phone numbers, residential address, salary information, etc. They are important because a data leak in these could put the employees at risk since threat actors could acquire them and making them vulnerable to phishing (or other scams).
2. **Cloud storage with the invoices**: If a threat actor were to acquire this information, it will put the customer in risk as they can use the invoice to scam the customers out of money which causes loss to the customer and damage to reputation for the company.
3. **Public Wi-Fi router**: If an attacker manages to gain access to the network, they can analyse the traffic, infect the systems with malicious software, steal information, etc.
4. **Company website**: Any damages to the website can disrupt the services of the company and depending on the scale of the disruption, it can cause large losses or damage to reputation for the company.
5. **Shared computers**: Damage to computers as an asset could disrupt the work of multiple employees.

THREE POTENTIAL THREATS

1. **Infected system introduced into the network**: A physical device like a mobile phone that is infected with malicious software could compromise the entire network. The threat actor could be an employee (willing/unwilling) or simply a visitor connecting to the network using the public Wi-Fi.
2. **Accessing the shared computers with malicious intent**: An employee could access accounts of another employee via the computer to gain confidential information, perform privilege escalation, etc.
3. **Cyber attacks on the website**: Threat actors could try to access the admin credentials, perform SQLi, disrupt the service using DDoS attacks, etc.

THREE VULNERABILITIES

1. **Use of public Wi-Fi**: Using public Wi-Fi is often less secure as intruders can snoop into the data traffic within the network. Especially in a company, this could be exploited for information gain and malicious access to internal resources.
2. **Computers without proper security**: Without proper authentication or authorization, anyone could access anyone else's account. This is risky since a lower-level employee could access sensitive information that is not meant for them.
3. **Hard coded and weak passwords in websites**: An attacker could access sensitive information like admin credentials if the passwords are weak or is stored in the code.

ASSOCIATED RISKS

1. **Information loss**: Insecure network makes the devices and the traffic vulnerable to data theft as the attackers can access information by snooping into the network.
2. **Account modification or deletion**: By gaining access to accounts of other employees, an employee with malicious intent can delete/modify files, data, accounts, etc.
3. **Disruption of services**: By accessing the admin credentials or performing cyber attacks against the website, the attacker can temporarily or permanently disrupt the services provided by the company for the legitimate users.

SECURITY RECOMMENDATIONS

1. Using a more **secure network** instead of public Wi-Fi to prevent data loss.
2. Securing the network with **strong passwords** and ensuring only employees can access the network.
3. Implementing a **Zero-Trust model** ensuring each interaction only has limited privileges, ensuring no employee can modify or edit the accounts of another employee.
4. Ensuring that passwords are stored separately and proper **secure hashing algorithms** are used on the website.
5. Using **secure cloud platforms** to ensure third-party personals cannot access sensitive information like invoices.